A Monomial Pandrosion Scheme
Contraction, Chart Scaling, and Steffensen Acceleration for $z^p = x$

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Abstract

This paper gives a self-contained, monomial-only account of the Pandrosion scheme for

$$z^p = x.$$

The purpose is not to compete with Newton's method as a production algorithm for ordinary floating-point root extraction. Newton is usually cheaper there. The point is different: the ancient proportional-mean geometry leads to a single derivative-free fixed-point map whose residual profile, scaling behavior, and chart changes can be analyzed exactly.

The central map is

$$h_{p,x}(s) = 1 - \frac{x-1}{x(1+s+\cdots+s^{p-1})}.$$

Its fixed point is $s_* = x^{-1/p}$, and the desired root is recovered by $v = xs_*^{p-1} = x^{1/p}$. The linear multiplier is explicit:

$$\lambda_{p,x} = \frac{(\alpha-1)\sum_{k=1}^{p-1} k\alpha^{p-1-k}}{\sum_{k=0}^{p-1}\alpha^k} = 1 - \frac{p}{S_p(\alpha)} = \frac{\alpha^p - p\alpha + p - 1}{\alpha^p - 1}, \qquad \alpha = x^{1/p},$$

where $S_p(\alpha) = 1 + \alpha + \cdots + \alpha^{p-1}$ and the rational expression is understood by its removable value at $\alpha = 1$. The compact form $1 - p/S_p(\alpha)$ is the first organizing fact of the paper: it explains why raw Pandrosion slows on large targets and why homothetic preconditioning ("Thales scaling") helps.

The second organizing fact is chart selection. For very small targets, the right geometry is not the original affine chart but the reciprocal chart of the Riemann sphere. In that chart one can scale again, run the same Pandrosion map near a normalized target $y \approx 1$, and return to the original variable. The local expansion

$$\lambda_{p,e^\delta} = \frac{p-1}{2p}\delta + O_p(\delta^2)$$

explains the first-order size of the residual factor, while strict monotonicity of $\lambda_{p,y}$ on $y \geq 1$ proves the global logarithmic rule in the admissible branch: choose the chart and homothety that minimize $|\log y|$. The main quantitative result is a sharp minimax scale-palette theorem. If available root-scales are $q^{\mathbb{Z}}$, then direct scaling alone has sharp worst-case multiplier $1 - p/S_p(q)$, whereas allowing the reciprocal chart improves the sharp worst-case bound to $1 - p/S_p(\sqrt{q})$, independently of the magnitude of x for fixed p and q. More generally, among all K-offset chart-scalings with the same logarithmic period, the best possible worst-case raw multiplier is

$$1 - \frac{p}{S_p(q^{1/(2K)})},$$

and it is attained by equally spaced logarithmic offsets. Thus chart scaling is not only a useful normalization rule, but an optimal one in a natural finite-palette model.

Finally, Aitken–Steffensen acceleration turns the linearly convergent map into a derivative-free quadratically convergent map, with an explicit local error constant. The complex monomial extension is kept at the level needed for an expository account: cyclotomic anchors, local seeding disks, and a finite selector. The result is a deliberately narrow but coherent account of the Pandrosion monomial scheme: fixed point, contraction, scaling, reciprocal charts, acceleration, and complex roots.

One-sentence summary. *Pandrosion supplies a geometric fixed-point map; its distinctive contribution here is the exact monomial residual profile and the rule that chart scaling should make $|\log y|$ small before iteration begins.*

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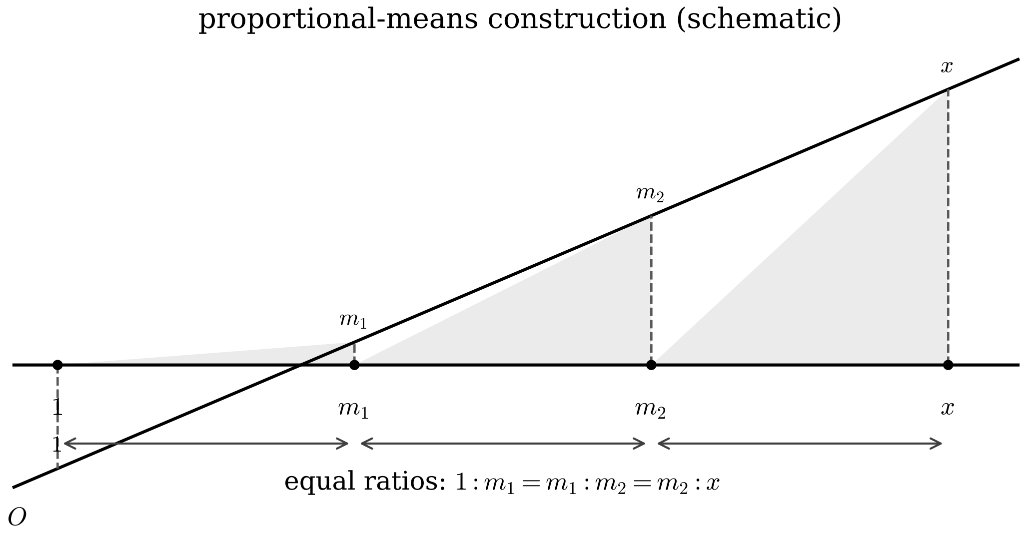

Figure 1: Proportional-means construction, with lengths not drawn to scale. The dashed parallels encode the Thales-type step, and the shaded triangles indicate the repeated similarity condition $1 : m_1 = m_1 : m_2 = m_2 : x$.

1 Introduction and roadmap

The equation $z^p = x$ is one of the simplest possible test cases for an iteration. It is simple enough that Newton's method, Halley's method, and the classical higher-order families are already extremely efficient. The interest of Pandrosion is therefore not numerical dominance. The interest is that a geometric proportional-means construction gives a fixed-point map whose multiplier can be computed exactly and whose behavior under homothety and reciprocal chart changes can be seen without hidden constants.

The paper proceeds in four steps. Sections 2–5 construct the map and compute its residual multiplier. Sections 10–12.1 explain why Thales scaling and Riemann-chart scaling are the same logarithmic preconditioning principle. The scale-lattice results then turn that principle into a sharp finite-palette bound. Finally, Sections 13–15 compare the method with standard iterations and record the complex monomial multi-start picture.

2 The cube-root construction

Pandrosion's construction begins with the ancient problem of doubling the cube [3]: constructing a segment of length $\sqrt[3]{2}$ from a unit segment. Algebraically, this is the problem of inserting two proportional means m_1, m_2 between 1 and 2:

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2}.$$

The solution is $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$.

Remark 2.1 (Historical caution). The name "Pandrosion" is used here as historical motivation for a family of proportional-mean constructions. The surviving evidence is indirect and passes through later sources such as Pappus [1, 2]. None of the convergence claims below rests on the historical attribution; they rest only on the explicit map (2) and its generalization.

Pandrosion's geometric procedure does not produce the exact value in finitely many steps. It produces a sequence of approximations. This is the key modern viewpoint: the construction is an iteration, and the error sequence of that iteration is its residual profile.

In the classical cube-root case one may write the construction with an internal length u_n and a visible readout v_n:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\frac{4-u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4-u_n}{4}\right)^2. \tag{1}$$

The number u_n belongs to the diagram. The number v_n is the approximation that is read from the diagram.

The substitution

$$s_n = \frac{4-u_n}{4}$$

removes the distracting constants. Since $u_n = 4(1 - s_n)$, the readout becomes $v_n = 2s_n^2$, and the iteration becomes

$$s_{n+1} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2} = 1 - \frac{1}{2(1 + s_n + s_n^2)}. \tag{2}$$

Proposition 2.2 (Cube-root fixed point)**.** *The map in (2) has the fixed point*

$$s_* = 2^{-1/3}.$$

Consequently the readout satisfies

$$v_n = 2s_n^2 \longrightarrow 2s_*^2 = \sqrt[3]{2}.$$

Proof. The fixed-point equation is

$$s = 1 - \frac{1}{2(1+s+s^2)}.$$

Multiplying by $2(1+s+s^2)$ gives

$$2s(1+s+s^2) = 2(1+s+s^2) - 1,$$

which simplifies to $2s^3 = 1$. Hence $s = 2^{-1/3}$ in the positive real branch. The readout identity follows immediately. □

The diagram computes $\sqrt[3]{2}$ indirectly. It first converges to $2^{-1/3}$, and the visible root is recovered by the readout $v = 2s^2$.

3 Why the polynomial S_p appears

The polynomial

$$S_p(s) = 1 + s + \cdots + s^{p-1}$$

appears for a geometric reason. Thales' theorem constructs proportional segments. Repeating the construction builds a finite geometric progression of ratios. Algebraically that progression is exactly S_p.

The elementary identity behind the construction is

$$1 - s^p = (1-s)(1+s+\cdots+s^{p-1}) = (1-s)S_p(s). \tag{3}$$

Thus, whenever a diagram compares 1 and s^p through successive proportional segments, the denominator $S_p(s)$ is the natural object that records the accumulated proportional means.

This identity is the bridge between the geometric construction and the fixed-point map.

4 The universal real-line iteration

Let $p \geq 2$ and $x > 0$. Define

$$S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad h_{p,x}(s) = 1 - \frac{x-1}{xS_p(s)}. \tag{4}$$

The Pandrosion iteration is

$$s_{n+1} = h_{p,x}(s_n). \tag{5}$$

The desired root is not s_n itself. It is read from s_n by

$$v_n = x s_n^{p-1}. \tag{6}$$

Theorem 4.1 (Fixed point and readout)**.** *Let $x > 1$ and $p \geq 2$. The map $h_{p,x}$ has the positive fixed point*

$$s_* = x^{-1/p}.$$

At this fixed point the readout is

$$x s_*^{p-1} = x^{1/p}.$$

Proof. The fixed-point equation $s = h_{p,x}(s)$ is

$$1 - s = \frac{x-1}{xS_p(s)}.$$

Using (3), this is equivalent to

$$x(1-s)S_p(s) = x - 1, \qquad \text{hence} \qquad x(1-s^p) = x-1.$$

Therefore $xs^p = 1$, so $s = x^{-1/p}$ in the positive branch. The readout is then $x(x^{-1/p})^{p-1} = x^{1/p}$. □

Remark 4.2. The map $h_{p,x}$ is derivative-free in its definition. It is built from the finite geometric sum S_p, not from the derivative of $s^p - x$.

5 The contraction ratio

The speed of the raw Pandrosion iteration is governed by the derivative of $h_{p,x}$ at the fixed point. This derivative is not used to define the method; it is used only to analyze the residual profile.

Let

$$\alpha = x^{1/p}.$$

Then $s_* = \alpha^{-1}$.

Theorem 5.1 (Closed-form contraction factor). *For $x > 1$ and $p \geq 2$, the linear contraction factor of the Pandrosion iteration at the fixed point is*

$$\lambda_{p,x} = h'_{p,x}(s_*) = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}. \tag{7}$$

Equivalently, with $S_p(\alpha) = 1 + \alpha + \cdots + \alpha^{p-1}$,

$$\lambda_{p,x} = 1 - \frac{p}{S_p(\alpha)} = \frac{\alpha^p - p\alpha + p - 1}{\alpha^p - 1}, \tag{8}$$

where the final quotient has removable value 0 at $\alpha = 1$. In particular,

$$\lambda_{2,x} = \frac{\alpha - 1}{\alpha + 1}, \qquad \lambda_{3,x} = \frac{(\alpha - 1)(\alpha + 2)}{\alpha^2 + \alpha + 1}.$$

Proof. Differentiate (4):

$$h'_{p,x}(s) = \frac{x-1}{x} \frac{S'_p(s)}{S_p(s)^2}.$$

Substitute $s = \alpha^{-1}$ and use

$$S_p(\alpha^{-1}) = \sum_{k=0}^{p-1} \alpha^{-k} = \alpha^{1-p} \sum_{k=0}^{p-1} \alpha^k.$$

The same change of powers in $S'_p(\alpha^{-1})$ yields (7). For the compact formula, telescope the numerator:

$$(\alpha - 1)\sum_{k=1}^{p-1} k\alpha^{p-1-k} = \sum_{k=0}^{p-1} \alpha^k - p = S_p(\alpha) - p.$$

Dividing by $S_p(\alpha)$ gives the first equality in (8); the rational quotient follows from $S_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$ for $\alpha \neq 1$. □

For later chart selection it is useful to record an equivalent one-line form of this multiplier.

Proposition 5.2 (Strict monotonicity of the monomial multiplier). *Fix $p \geq 2$ and put $y = e^\delta \geq 1$. Then*

$$\delta \longmapsto \lambda_{p,e^\delta}$$

is continuous on $[0, \infty)$, strictly increasing on $(0, \infty)$, satisfies $\lambda_{p,1} = 0$, and tends to 1 as $\delta \to \infty$. Equivalently, $y \mapsto \lambda_{p,y}$ is strictly increasing on $[1, \infty)$.

Proof. Let

$$\alpha = y^{1/p} = e^{\delta/p}.$$

By the compact formula (8), for $\alpha > 1$,

$$\lambda_{p,y} = \frac{\alpha^p - p\alpha + p - 1}{\alpha^p - 1}.$$

Differentiate that rational function with respect to α:

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\lambda_{p,y} = \frac{p\big((p-1)\alpha^p - p\alpha^{p-1} + 1\big)}{(\alpha^p - 1)^2}.$$

It remains only to check the sign of

$$g_p(\alpha) = (p-1)\alpha^p - p\alpha^{p-1} + 1.$$

We have $g_p(1) = 0$ and

$$g'_p(\alpha) = p(p-1)\alpha^{p-2}(\alpha - 1) > 0 \qquad (\alpha > 1).$$

Therefore $g_p(\alpha) > 0$ for every $\alpha > 1$, hence $\lambda_{p,y}$ is strictly increasing in α, and therefore also in $y = \alpha^p$ and in $\delta = p\log\alpha$. The endpoint value follows from the removable limit in (8), and the limit at infinity is immediate from the leading terms. □

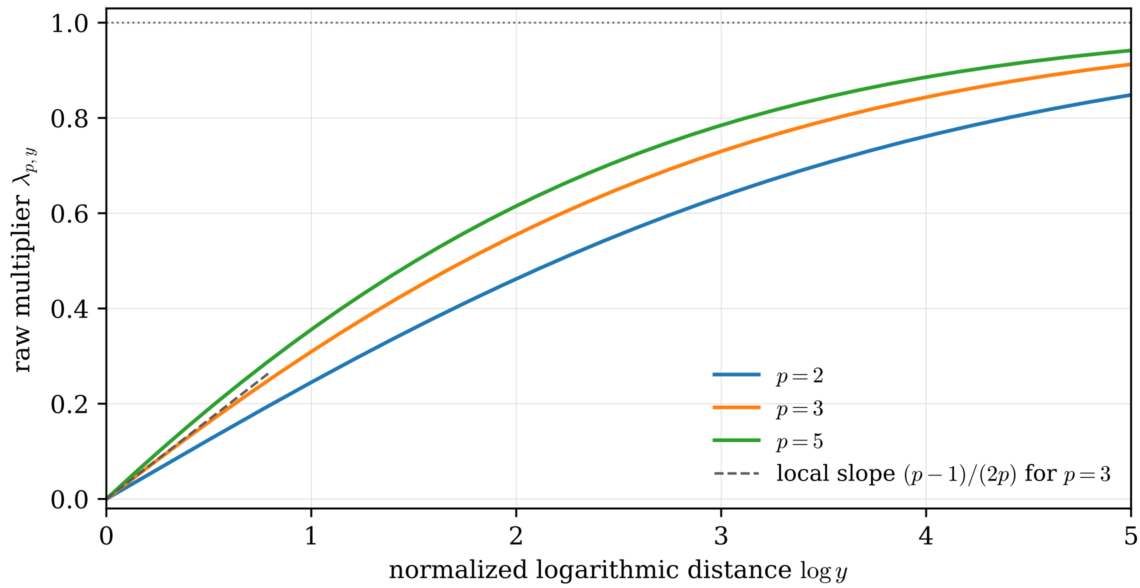

Figure 2: The residual fingerprint $\lambda_{p,y}$ as a function of the normalized logarithmic distance $\log y$. The curves are monotone on the admissible branch $y \geq 1$, and their initial slope is $(p-1)/(2p)$.

The formula immediately explains the qualitative behavior:

- if x is close to 1, then $\lambda_{p,x}$ is small and the raw geometry converges quickly;
- if x is large, then $\lambda_{p,x}$ approaches 1 and the raw geometry converges slowly;
- increasing p at fixed large target makes the unscaled construction more delicate.

Proposition 5.3 (Large-p limit at fixed target)**.** *For fixed $x > 1$,*

$$\lim_{p \to \infty} \lambda_{p,x} = 1 - \frac{\log x}{x-1}.$$

Proof. Put $L = \log x > 0$, so that $\alpha = e^{L/p}$. Use the compact formula

$$\lambda_{p,x} = 1 - \frac{p}{S_p(e^{L/p})}.$$

The only limit needed is the Riemann sum

$$\frac{1}{p} S_p(e^{L/p}) = \frac{1}{p} \sum_{k=0}^{p-1} e^{Lk/p} \longrightarrow \int_0^1 e^{Lt}\, \mathrm{d}t = \frac{e^L - 1}{L} = \frac{x-1}{\log x}.$$

Therefore

$$\frac{p}{S_p(e^{L/p})} \longrightarrow \frac{\log x}{x-1},$$

and the claimed limit follows. This proof avoids any interchange of a growing weighted numerator with an error term; it uses only the uniform continuity of $t \mapsto e^{Lt}$ on $[0, 1]$. □

The number $\lambda_{p,x}$ is the fingerprint of Pandrosion's residual profile. Scaling works because it changes x before this fingerprint is formed.

Remark 5.4 (Choosing a scale from a contraction threshold). The simplified form (8) also gives an explicit scale selection test. Fix a desired raw multiplier $0 < \tau < 1$. Since $\lambda_{p,y}$ is continuous and strictly increasing on $y \geq 1$, there is a unique $\alpha_\tau > 1$ satisfying

$$\frac{\alpha_\tau^p - p\alpha_\tau + p - 1}{\alpha_\tau^p - 1} = \tau.$$

Thus every normalized target with

$$1 \leq y \leq \alpha_\tau^p$$

has raw multiplier at most τ. In a discrete scale family, one simply chooses the available homothety whose normalized target lies closest to 1; Corollary 12.4 below says that this also minimizes the exact multiplier among the available admissible choices.

6 Error bounds and iteration counts

The local statement is the usual one for a contracting fixed-point map. If an interval I contains s_* and

$$\sup_{s\in I} |h'_{p,x}(s)| \le \Lambda < 1,$$

then every orbit that remains in I satisfies

$$|s_n - s_*| \le \Lambda^n |s_0 - s_*|. \tag{9}$$

Consequently the readout error is controlled by the same geometric decay, with an additional Lipschitz constant for $s \mapsto xs^{p-1}$ on I.

A practical iteration budget follows from (9). To reach $|s_n - s_*| \le \tau$, it is enough to take

$$n \ge \frac{\log(\tau/|s_0 - s_*|)}{\log \Lambda}. \tag{10}$$

The denominator is negative, so a smaller Λ means a smaller budget. This is the numerical meaning of the geometric statement: fewer effective contractions are needed when the normalized multiplier is smaller.

7 The case $0 < x < 1$

When $0 < x < 1$, the fixed point

$$s_* = x^{-1/p}$$

lies outside $(0, 1)$. This is not only a cosmetic inconvenience. It changes the real dynamics of the direct map.

Indeed, for every $s > 0$ one has $S_p(s) > 0$, and therefore

$$h_{p,x}(s) = 1 + \frac{1-x}{xS_p(s)} > 1.$$

Thus even if the diagram is initialized inside $(0, 1)$, the next internal point jumps outside that interval. Moreover

$$h'_{p,x}(s) = \frac{x-1}{x} \frac{S'_p(s)}{S_p(s)^2} < 0 \qquad (s > 0),$$

so the map is orientation-reversing on the positive real line. At the fixed point, writing $\rho = s_* = x^{-1/p} > 1$, the direct multiplier is

$$h'_{p,x}(s_*) = -\mu_{p,x}, \qquad \mu_{p,x} = \frac{(\rho-1)S'_p(\rho)}{S_p(\rho)}. \tag{11}$$

This is not the same as the positive-target multiplier of $1/x$. In fact, as $x \to 0^+$ one has $\rho \to \infty$ and

$$S_p(\rho) = \rho^{p-1} + O(\rho^{p-2}), \qquad S'_p(\rho) = (p-1)\rho^{p-2} + O(\rho^{p-3}),$$

so

$$\mu_{p,x} = \frac{(\rho-1)S'_p(\rho)}{S_p(\rho)} \longrightarrow p-1.$$

Hence for $p \ge 3$ the direct chart is locally unstable for sufficiently small x. Even when the direct map happens to converge, it no longer has the same clean "nested proportional means inside a unit interval" interpretation used for $x > 1$.

The canonical start also shows the scale problem clearly:

$$h_{p,x}(1) = 1 + \frac{1-x}{xp}.$$

For very small x, this point is already very large. In other words, the direct affine chart asks the construction to travel to the far side of the positive line before it can read the root.

The practical resolution is the reciprocal target. Since

$$x^{1/p} = \frac{1}{(1/x)^{1/p}},$$

one applies the $x > 1$ theory to $1/x$ and then inverts the result. This restores the stable multiplier $\lambda_{p,1/x} < 1$ from Theorem 5.1 instead of the direct orientation-reversing multiplier (11). In the reciprocal target $y = 1/x > 1$, the internal fixed point is

$$s_y = y^{-1/p} = x^{1/p},$$

which is already the desired root of the original target. This is the first appearance of a principle that becomes central later: changing charts is often better than forcing one affine chart to handle every scale.

For $0 < x < 1$, direct Pandrosion is not algebraically wrong, but it is geometrically misplaced. The reciprocal chart restores the same positive-target geometry while keeping the normalized fixed point near the intended root.

8 A better starting point

The raw iteration needs an initial point. The most natural choice is not always the fastest. A simple geometric improvement is to start from the image of the neutral point 1:

$$s_0^{\mathrm{opt}} = h_{p,x}(1) = 1 - \frac{x-1}{xp}. \tag{12}$$

This point already incorporates one full proportional-mean step. In the language of the diagram, one avoids spending future parallels to do work that can be done once at initialization.

Proposition 8.1 (Canonical start)**.** *The start $s_0^{\mathrm{opt}} = h_{p,x}(1)$ is the canonical one-step start produced by the same Pandrosion geometry. Near $x = 1$ it satisfies*

$$s_0^{\mathrm{opt}} = x^{-1/p} + O((x-1)^2).$$

Proof. The first statement is the definition. The expansion follows from $x^{-1/p} = 1-(x-1)/p+O((x-1)^2)$ and from $1-(x-1)/(xp) = 1-(x-1)/p+O((x-1)^2)$. □

Thus the canonical start removes the first-order local error of the neutral start $s = 1$. Starting from 1 gives $s_* - 1 = -(x-1)/p + O((x-1)^2)$, whereas starting from $h_{p,x}(1)$ leaves only an $O((x-1)^2)$ discrepancy. This is the small but useful quantitative reason why the numerical counts below always start from the canonical value.

Starting-point optimization is not an extra algorithmic layer. It is the first Pandrosion step used deliberately, before the iteration budget begins.

9 Steffensen acceleration

The raw Pandrosion map is geometrically meaningful but only linearly convergent. Aitken–Steffensen acceleration turns any sufficiently regular linearly convergent fixed-point iteration into a quadratically convergent one, without requiring derivatives [5, 4, 6].

For $h = h_{p,x}$, define

$$\sigma_{p,x}(s) = s - \frac{(h(s)-s)^2}{h(h(s)) - 2h(s) + s}. \tag{13}$$

Theorem 9.1 (Quadratic acceleration with explicit coefficient)**.** *Assume h is analytic near s_*, $h(s_*) = s_*$, and*

$$\lambda = h'(s_*) \neq 1.$$

Then the Steffensen map $\sigma_{p,x}$ has a removable singularity at s_*, fixes s_*, and satisfies

$$\sigma_{p,x}(s) - s_* = \frac{\lambda h''(s_*)}{2(\lambda-1)}(s-s_*)^2 + O((s-s_*)^3). \tag{14}$$

In particular, if $\lambda h''(s_*) \neq 0$, the convergence is genuinely quadratic. If $\lambda = 0$, or if $h''(s_*) = 0$, the displayed quadratic coefficient vanishes and the acceleration is at least quadratic, often of higher local order.

Proof. Write $e = s - s_*$ and expand

$$h(s_* + e) = s_* + \lambda e + ae^2 + be^3 + O(e^4), \qquad a = \frac{1}{2}h''(s_*).$$

Then

$$h(s) - s = (\lambda - 1)e + ae^2 + be^3 + O(e^4).$$

A second substitution gives

$$h(h(s_* + e)) = s_* + \lambda^2 e + a\lambda(1+\lambda)e^2 + O(e^3).$$

Hence the denominator in (13) is

$$h(h(s)) - 2h(s) + s = (\lambda-1)^2 e + a(\lambda-1)(\lambda+2)e^2 + O(e^3).$$

Since $\lambda \neq 1$, this denominator has a simple zero at $e = 0$ and the quotient in (13) has a removable singularity. Dividing the two series,

$$\frac{(h(s)-s)^2}{h(h(s)) - 2h(s) + s} = e - \frac{a\lambda}{\lambda-1}e^2 + O(e^3).$$

Therefore

$$\sigma_{p,x}(s) - s_* = e - \left(e - \frac{a\lambda}{\lambda-1}e^2 + O(e^3)\right) = \frac{a\lambda}{\lambda-1}e^2 + O(e^3),$$

which is (14). □

For the Pandrosion map itself,

$$h''_{p,x}(s) = \frac{x-1}{x} \frac{S''_p(s)S_p(s) - 2(S'_p(s))^2}{S_p(s)^3}.$$

This second derivative is non-zero throughout the useful positive real regime $x > 1$. Indeed, for $s > 0$,

$$2(S'_p(s))^2 - S_p(s)S''_p(s) = ps^{p-2} \sum_{r=0}^{p-2} (r+1)(p-r-1)s^r > 0. \tag{15}$$

For completeness, here is the coefficient check. Write $N = p - 1$. The coefficient of s^m in $2(S'_p)^2 - S_p S''_p$ is 0 for $0 \le m < N - 1$, because

$$2\sum_{i=1}^{m+1} i(m+2-i) - \sum_{i=2}^{m+2} i(i-1) = 0.$$

For $m = N - 1 + r$, $0 \le r \le N - 1$, the same coefficient is

$$2\sum_{i=r+1}^{N} i(N+1+r-i) - \sum_{i=\max(2,r+1)}^{N} i(i-1) = (N+1)(r+1)(N-r),$$

which is exactly the coefficient of the right-hand side of (15). Thus the displayed identity is an elementary finite-sum identity, not an additional analytic assumption. Thus, when $x > 1$,

$$h''_{p,x}(s) = -\frac{x-1}{x} \frac{2(S'_p(s))^2 - S_p(s)S''_p(s)}{S_p(s)^3} < 0.$$

Combined with $0 < \lambda_{p,x} < 1$, this proves that Steffensen–Pandrosion is genuinely quadratic for every $p \ge 2$ and $x > 1$.

Steffensen acceleration is especially natural here. The underlying geometric map remains derivative-free, and the acceleration uses only evaluations of the same map.

10 Thales scaling

Scaling is the most important practical improvement. Suppose x is large. Instead of applying Pandrosion directly to x, write

$$x = Ax', \tag{16}$$

where A is chosen so that x' is close to 1. Then

$$x^{1/p} = A^{1/p}(x')^{1/p}.$$

One computes $(x')^{1/p}$ by Pandrosion and multiplies back by $A^{1/p}$.

Equivalently, write $A = B^p$ and set $z = Bu$. The equation $z^p = x$ becomes

$$u^p = \frac{x}{B^p}.$$

Thus Thales scaling is a homothety in the root coordinate: the computation is performed on the normalized target x/B^p, and the result is transported back by the return map $u \mapsto Bu$.

The point is not cosmetic. The contraction factor is controlled by the normalized target, not by the original target. Since $\lambda_{p,x}$ becomes small when x is close to 1, the scaled problem may require far fewer effective Thales constructions.

> *Homothety does not merely normalize the input. It changes the geometry before Pandrosion begins. A good homothety reduces the number of proportional layers the construction must build.*

This is the one-dimensional ancestor of the chart-scaled construction developed next. Thales scaling normalizes the size of the problem inside one fixed affine chart. But a fixed chart may be the wrong place to measure the problem: near 0, scaling alone is less natural than first moving the computation to the reciprocal chart. The next step is therefore not a new iteration, but the same homothetic idea after a change of chart.

11 Geometric inversion

The reciprocal trick in Section 7 and the homothety of Section 10 are two faces of the same geometric principle:

> *choose a chart, normalize by homothety, run Pandrosion near the fixed point* 1*, and transport the answer back.*

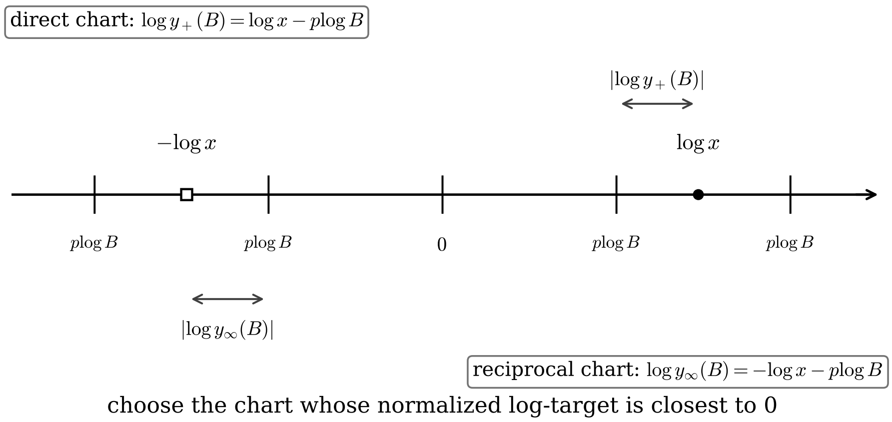

Figure 3: Direct and reciprocal logarithmic charts. The affine chart uses $\log y_+(B) = \log x - p\log B$, while the reciprocal chart uses $\log y_\infty(B) = -\log x - p\log B$. Chart selection chooses the line whose normalized logarithmic target lies closest to 0.

The reciprocal trick is not merely an algebraic convenience; it is a change of chart on the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The inversion

$$I(z) = \frac{1}{z} \tag{17}$$

exchanges the neighborhoods of 0 and ∞. Thus a target x that is small in the usual affine chart becomes the large target $1/x$ in the reciprocal chart.

For the equation

$$z^p = x,$$

set $w = I(z) = 1/z$. Then

$$w^p = \frac{1}{x}. \tag{18}$$

Therefore the root can be computed in the reciprocal coordinate and brought back by another inversion:

$$z = \frac{1}{w}.$$

This has a direct Pandrosion interpretation. If $0 < x < 1$ and $y = 1/x > 1$, then Pandrosion applied to y has internal fixed point

$$s_y = y^{-1/p} = x^{1/p}.$$

So, in the reciprocal chart, the fixed point itself is already the desired root of the original target x. If one uses the usual readout instead, one obtains

$$y s_y^{p-1} = y^{1/p} = x^{-1/p},$$

and the final inversion gives $x^{1/p}$.

Proposition 11.1 (Correctness of reciprocal Pandrosion)**.** *Let $0 < x < 1$, $y = 1/x$, and suppose the Pandrosion readout for y converges to $y^{1/p}$. Then*

$$\frac{1}{y^{1/p}} = x^{1/p}.$$

Equivalently, the internal fixed point $s_y = y^{-1/p}$ is already equal to $x^{1/p}$.

Proof. The first identity is immediate from $y = 1/x$. The second follows from $s_y = y^{-1/p} = (1/x)^{-1/p} = x^{1/p}$. □

The interval $0 < x < 1$ is not a failure of Pandrosion. It asks for a different chart. On the Riemann sphere, inversion moves the problem away from the small scale near 0 and into the same positive-target geometry used for $x > 1$.

12 Inversion, homothety, and return of chart

Inversion alone makes the interval $0 < x < 1$ compatible with the $x > 1$ theory, while Thales scaling makes large targets smaller. The useful acceleration is to combine both operations. If x is very small, then $1/x$ is very large, and the contraction factor $\lambda_{p,1/x}$ can be close to 1. The accelerated geometric idea is therefore:

invert, apply a homothety in the reciprocal chart, run Pandrosion near 1, and then return to the original chart.

Let $B > 0$ be a scale factor in the root coordinate. There are two natural charts.

Direct affine chart. Set

$$z = Bu.$$

Then $z^p = x$ is equivalent to

$$u^p = y_+(B), \qquad y_+(B) = \frac{x}{B^p}. \tag{19}$$

After computing $u \approx y_+(B)^{1/p}$, return

$$z \approx Bu.$$

Inverted chart. Set

$$z = \frac{1}{Bu}.$$

Then $z^p = x$ is equivalent to

$$u^p = y_\infty(B), \qquad y_\infty(B) = \frac{1}{xB^p}. \tag{20}$$

After computing $u \approx y_\infty(B)^{1/p}$, return

$$z \approx \frac{1}{Bu}.$$

Proposition 12.1 (Chart-scaled correctness)**.** *Let $B > 0$.*

1. *If $u^p = x/B^p$, then $(Bu)^p = x$.*
2. *If $u^p = 1/(xB^p)$, then $(1/(Bu))^p = x$.*

Thus both the direct chart and the inverted chart solve the original equation after their respective return maps.

Proof. The first identity gives $(Bu)^p = B^p u^p = B^p(x/B^p) = x$. The second gives

$$\left(\frac{1}{Bu}\right)^p = \frac{1}{B^p u^p} = \frac{1}{B^p/(xB^p)} = x.$$

□

The speed is governed by the normalized target, not by the original target. In the direct chart the relevant contraction factor is

$$\lambda_{p,y_+(B)},$$

whereas in the inverted chart it is

$$\lambda_{p,y_\infty(B)}.$$

Thus one should choose the chart and the scale so that the normalized target is as close to 1 as possible. This is why the logarithmic distance appears below. Since $|\log y|$ measures multiplicative distance from 1, minimizing it is a scale-invariant way to move the internal fixed point $y^{-1/p}$ toward 1. By the contraction analysis of Section 5, this is exactly the region where raw Pandrosion has its smallest linear residual factor. A concise selection rule is

$$(\chi, B) \in \operatorname*{argmin}_{\chi \in \{+,\infty\},\, B \in \mathcal{B}} |\log y_\chi(B)|, \tag{21}$$

with the admissibility constraint that the chosen normalized target lies in the positive branch where the monotonicity theorem applies, namely $y_\chi(B) \geq 1$. Here $\mathcal{B}$ is the available family of homothetic scales: powers of 10, powers of 2, or any constructible scale family allowed by the geometric model.

Remark 12.2. Equation (21) is the Riemann-sphere analogue of Thales scaling. Ordinary scaling chooses the size of the affine chart. Riemann-chart scaling also chooses whether the computation should be performed near 0 or near ∞. Once that chart is selected, nothing else changes: the same Pandrosion map is applied to the normalized target, and the return map carries the result back to the original variable.

12.1 A local contraction lemma for chart selection

The rule (21) can be justified directly from the closed contraction formula. In fact, Proposition 5.2 proves that, once the admissible normalized targets satisfy $y \geq 1$, minimizing $\log y$ is exactly the same thing as minimizing the raw linear multiplier $\lambda_{p,y}$. The local expansion below explains the first-order size of that multiplier near the ideal target 1.

Write a normalized positive target as

$$y = e^\delta, \qquad \delta = \log y.$$

In the Pandrosion regime used here one usually arranges $y \geq 1$, so $\delta \geq 0$. The ideal scaled target is $y = 1$, for which the root in the normalized coordinate is exactly 1 and the residual geometry disappears. The following expansion says that targets near 1 have a contraction factor proportional to their logarithmic distance from 1.

Lemma 12.3 (Local contraction factor near the normalized target)**.** *For fixed $p \geq 2$ and $y = e^\delta$ with $\delta \geq 0$,*

$$\lambda_{p,e^\delta} = \frac{p-1}{2p}\,\delta + O_p(\delta^2) \qquad (\delta \to 0^+).$$

Equivalently,

$$\lambda_{p,y} = \frac{p-1}{2p}\,\log y + O_p((\log y)^2) \qquad (y \to 1^+).$$

Proof. Put $\alpha = y^{1/p} = e^{\delta/p}$. Uniformly for the finitely many exponents that occur below,

$$\alpha^j = e^{j\delta/p} = 1 + \frac{j\delta}{p} + O_p(\delta^2), \qquad \alpha - 1 = \frac{\delta}{p} + O_p(\delta^2).$$

Therefore

$$\sum_{k=0}^{p-1} \alpha^k = p + \frac{\delta}{p}\sum_{k=0}^{p-1} k + O_p(\delta^2) = p + \frac{p-1}{2}\delta + O_p(\delta^2),$$

and

$$\sum_{k=1}^{p-1} k\alpha^{p-1-k} = \sum_{k=1}^{p-1} k + \frac{\delta}{p}\sum_{k=1}^{p-1} k(p-1-k) + O_p(\delta^2) = \frac{p(p-1)}{2} + O_p(\delta).$$

Substituting these two estimates in (7) gives

$$\lambda_{p,e^\delta} = \frac{\left(\frac{\delta}{p} + O_p(\delta^2)\right)\left(\frac{p(p-1)}{2} + O_p(\delta)\right)}{p + O_p(\delta)} = \frac{p-1}{2p}\delta + O_p(\delta^2),$$

which is the announced expansion. □

Corollary 12.4 (Exact logarithmic chart rule in the admissible regime)**.** *Let $\mathcal{Y}$ be any finite or infinite family of admissible normalized targets with $y \geq 1$. For fixed $p \geq 2$, a target $y_* \in \mathcal{Y}$ minimizes $\log y$ over $\mathcal{Y}$ if and only if it minimizes the exact linear residual factor $\lambda_{p,y}$ over $\mathcal{Y}$.*

In particular, if the allowed homotheties form a discrete scale family, such as powers of 2 or powers of 10, then the best available logarithmic approximation to 1 is also the best available choice for the raw Pandrosion multiplier.

Proof. On the admissible branch $y \geq 1$, $|\log y| = \log y$, and the logarithm is strictly increasing in y. Proposition 5.2 says that $\lambda_{p,y}$ is strictly increasing in the same variable. Therefore the two minimization problems have exactly the same minimizers. □

The same statement can be read directly in the scale parameter. In the direct chart,

$$\log y_+(B) = \log x - p \log B,$$

whereas in the reciprocal chart,

$$\log y_\infty(B) = -\log x - p \log B.$$

Thus chart selection asks for a scale B whose logarithmic size best cancels $\log x$ in whichever chart is being used. If the scale family is discrete, for example powers of 2 or powers of 10, the best available scale is the nearest logarithmic approximation. The monotonicity result above shows that this is not merely a Taylor heuristic: within the admissible branch $y \geq 1$, it minimizes the exact raw multiplier among all available chart-scaled candidates. This is the theoretical reason why the benchmark table below improves sharply: the normalized target is moved from a large value to a value whose logarithmic distance from 1 is small.

Theorem 12.5 (Tight two-chart scale-lattice bound)**.** *Fix $p \geq 2$ and a scale ratio $q > 1$. Let the available root-scales be the q-lattice*

$$\mathcal{B}_q = \{q^m : m \in \mathbb{Z}\}.$$

For $x > 0$, write $a = \log_q x$ and let

$$r = a - p\left\lfloor \frac{a}{p} \right\rfloor \in [0, p)$$

be the remainder of a modulo p. In the direct chart, the best admissible q-scale gives the normalized target q^r. In the reciprocal chart, the best admissible q-scale gives the normalized target q^{p-r} when $r > 0$, and 1 when $r = 0$. Therefore the optimal two-chart target is

$$y_*(x) = q^{r_*}, \qquad r_* = \begin{cases} 0, & r = 0, \\ \min(r, p-r), & 0 < r < p, \end{cases}$$

and the optimal raw Pandrosion multiplier among all admissible candidates is

$$\lambda_{p,y_*(x)}.$$

In particular,

$$\lambda_{p,y_*(x)} \le \lambda_{p,q^{p/2}} = 1 - \frac{p}{S_p(\sqrt{q})}. \tag{22}$$

The bound is sharp for this scale lattice and these two charts: equality is attained by targets with $r = p/2$.

Proof. In the direct chart one has

$$\log_q y_+(q^m) = a - pm.$$

The admissible condition $y_+(q^m) \ge 1$ is $a - pm \ge 0$, so the closest admissible choice to 1 is $m = \lfloor a/p \rfloor$, giving $y_+ = q^r$. Similarly, in the reciprocal chart,

$$\log_q y_\infty(q^m) = -a - pm.$$

The closest admissible choice is $m = \lfloor -a/p \rfloor$. The corresponding remainder is 0 if $r = 0$ and $p - r$ otherwise. When $r = 0$, the normalized target is exactly $y_* = 1$, and the removable value in (8) gives $\lambda_{p,1} = 0$. Hence the best two-chart logarithmic distance is $r_* \log q$.

By Corollary 12.4, minimizing this logarithmic distance in the admissible branch is exactly the same as minimizing the raw multiplier. Since $0 \le r_* \le p/2$, Proposition 5.2 gives

$$\lambda_{p,q^{r_*}} \le \lambda_{p,q^{p/2}}.$$

Finally, $q^{p/2} = (\sqrt{q})^p$, so the compact formula (8) gives

$$\lambda_{p,q^{p/2}} = 1 - \frac{p}{S_p(\sqrt{q})}.$$

If $r = p/2$, the direct chart gives the remainder $p/2$, while the reciprocal chart gives the complementary remainder $p - r = p/2$. Thus the two possible orientations meet at the same midpoint of the logarithmic cell. No admissible candidate in the two-chart q-lattice can have smaller logarithmic distance than $(p/2)\log q$, and the same monotonicity then proves sharpness of the multiplier bound. □

Corollary 12.6 (Magnitude-free local iteration budget)**.** *Under the hypotheses of Theorem 12.5, set*

$$\Lambda^{(2)}_{p,q} = 1 - \frac{p}{S_p(\sqrt{q})}.$$

After optimal two-chart q-lattice scaling, the asymptotic raw Pandrosion multiplier is bounded by $\Lambda^{(2)}_{p,q}$, independently of the magnitude of x for fixed p and q. Consequently, inside any local contraction interval, reducing an internal error by a factor $R > 1$ requires at most

$$\left\lceil \frac{\log R}{-\log \Lambda^{(2)}_{p,q}} \right\rceil$$

raw updates, up to the usual local-constant enlargement from Section 6. This worst-case bound is tight for the chosen q-lattice and the two charts. If $\Lambda^{(2)}_{p,q} = 0$, this means that the normalized target is exactly 1; equivalently, x lies exactly on the selected scale lattice, and the linearized residual disappears in one idealized step.

For comparison, direct scaling alone on the same q-lattice gives the sharp worst-case multiplier

$$\lambda_{p,q^p} = 1 - \frac{p}{S_p(q)}.$$

Thus the reciprocal chart halves the worst-case logarithmic lattice gap: direct scaling has $0 \le r < p$, while the two-chart rule has $0 \le r_* \le p/2$. The preceding theorem is the $K = 1$ case of a sharper minimax statement. It is useful because it separates what belongs to the monomial multiplier from what belongs to the logarithmic preconditioner itself. The combined table below records the direct-lattice bound and the optimal K-phase bounds in one place.

Theorem 12.7 (Minimax optimality of logarithmic chart palettes)**.** *Fix a logarithmic period $A = p \log q$ and let $K \ge 1$. Among all choices of K phase offsets on the circle $\mathbb{R}/A\mathbb{Z}$, the smallest possible worst-case logarithmic gap is*

$$\frac{A}{2K}.$$

It is attained if and only if the phases are equally spaced modulo A. Consequently, for monomial Pandrosion the sharp minimax raw multiplier for a K-phase logarithmic palette is

$$\lambda_{p,\exp(A/(2K))} = 1 - \frac{p}{S_p(q^{1/(2K)})}. \tag{23}$$

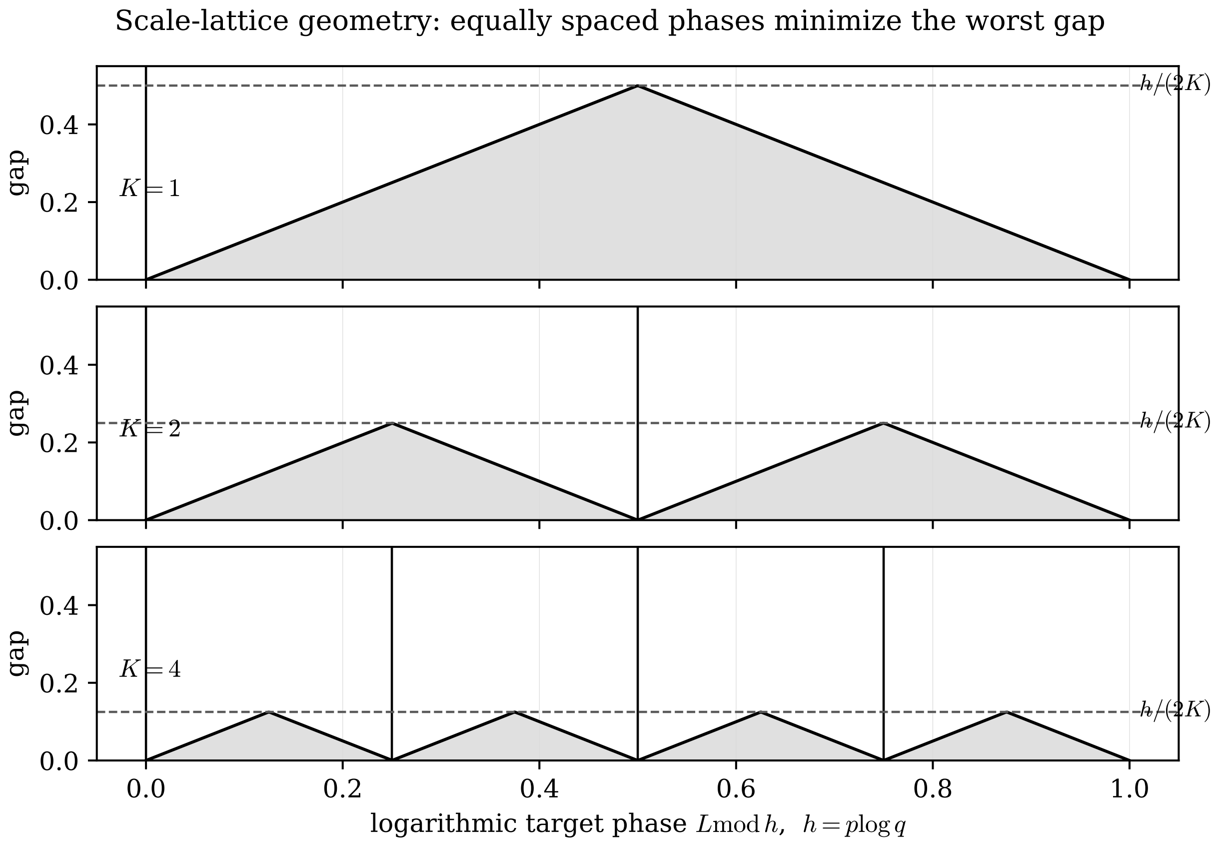

Figure 4: The scale-lattice theorem in one picture. For a logarithmic period $h = p\log q$, equally spaced K-phase palettes have covering radius $h/(2K)$, and no K-point palette can do better.

Proof. Put the K phase offsets on a circle of circumference A. The K gaps between consecutive offsets have total length A, so some gap has length at least A/K. Its midpoint is at distance at least $A/(2K)$ from every offset. This proves the lower bound. If the offsets are equally spaced, every gap has length A/K, hence every point of the circle lies within $A/(2K)$ of an offset. This proves optimality and equality. Since $\lambda_{p,y}$ is strictly increasing with $\log y$ on the admissible branch, the worst-case multiplier is minimized by the same covering-radius minimization. Finally, $\exp(A/(2K)) = q^{p/(2K)} = (q^{1/(2K)})^p$, so the compact formula (8) gives (23). □

Remark 12.8. The case $K = 1$ is the original two-chart $q^{\mathbb{Z}}$ lattice. Larger K corresponds to allowing a small finite list of precomputed offsets $B = q^{m+j/K}$. The theorem is mainly included to show that the logarithmic chart rule is not only sharp for the basic lattice: it is minimax optimal among finite periodic scale palettes.

q	p	direct	$K=1$	$K=2$	$K=4$
10	3	0.972973	0.788170	0.494997	0.270393
10	5	0.999550	0.965703	0.768132	0.481621
2	3	0.571429	0.320377	0.167458	0.085286
2	5	0.838710	0.555265	0.313678	0.165382
1.25	3	0.213115	0.109273	0.055239	0.027760
1.25	5	0.390766	0.209870	0.108350	0.054994

The "direct" column is $1 - p/S_p(q)$. The columns $K = 1, 2, 4$ display the sharp minimax multipliers $1 - p/S_p(q^{1/(2K)})$. The $K = 1$ column is exactly the two-chart bound of Theorem 12.5; the later columns show how precomputed offsets shrink the optimal logarithmic covering radius.

Thales scaling and Riemann-chart scaling are not separate tricks. Both are instances of the same contraction principle: make $|\log y|$ small before the Pandrosion map begins. Locally, $\lambda_{p,y} \sim ((p-1)/(2p)) \log y$; globally on $y \geq 1$, the exact multiplier is strictly increasing with $\log y$.

This logarithmic rule is the main conceptual novelty of the monomial account. The inversion $z \mapsto 1/z$ and the homothety $z \mapsto Bz$ are classical operations, but the closed multiplier (7) explains exactly why one should combine them: they are preconditioners for the residual factor, not post-processing tricks applied after convergence.

Numerical check. The following table was generated with high-precision Python arithmetic. All iterations are performed in the normalized chart named in the "chart" column before the final return map. In that column, *recip.*

means reciprocal chart only, while *recip.+scale* means reciprocal chart plus a homothety by B in the inverted root coordinate. Each row stops when the returned value has relative error at most 10^{-12} as an approximation to the normalized root $y^{1/p}$.

The column "raw" counts unaccelerated Pandrosion updates after the canonical start $s_0 = h_{p,y}(1)$. The column "Steff." counts Steffensen–Pandrosion updates from the same internal start. The column "Newton" counts Newton updates for $u^p = y$, started from the same visible canonical value

$$u_0 = y\, h_{p,y}(1)^{p-1}.$$

Thus the table reports both the Pandrosion phenomenon and the standard numerical baseline in exactly the same normalized problem.

p	x	chart	B	normalized y	$\lambda_{p,y}$	raw	Steff.	Newton	raw/Steff.
3	$2 \cdot 10^{-3}$	recip.	1	500	0.958294558	659	7	13	94
3	$2 \cdot 10^{-3}$	recip.+scale	5	4	0.412598948	30	3	5	10
3	$2 \cdot 10^{-6}$	recip.	1	500000	0.999529779	59506	13	24	4577
3	$2 \cdot 10^{-6}$	recip.+scale	50	4	0.412598948	30	3	5	10
5	$1/(3 \cdot 4^5)$	recip.	1	3072	0.993515266	4295	9	29	477
5	$1/(3 \cdot 4^5)$	recip.+scale	4	3	0.385672651	28	3	6	9.3
5	$5 \cdot 10^{-6}$	recip.	1	200000	0.999737824	106226	13	44	8171
5	$5 \cdot 10^{-6}$	recip.+scale	10	2	0.256508225	19	3	5	6.3

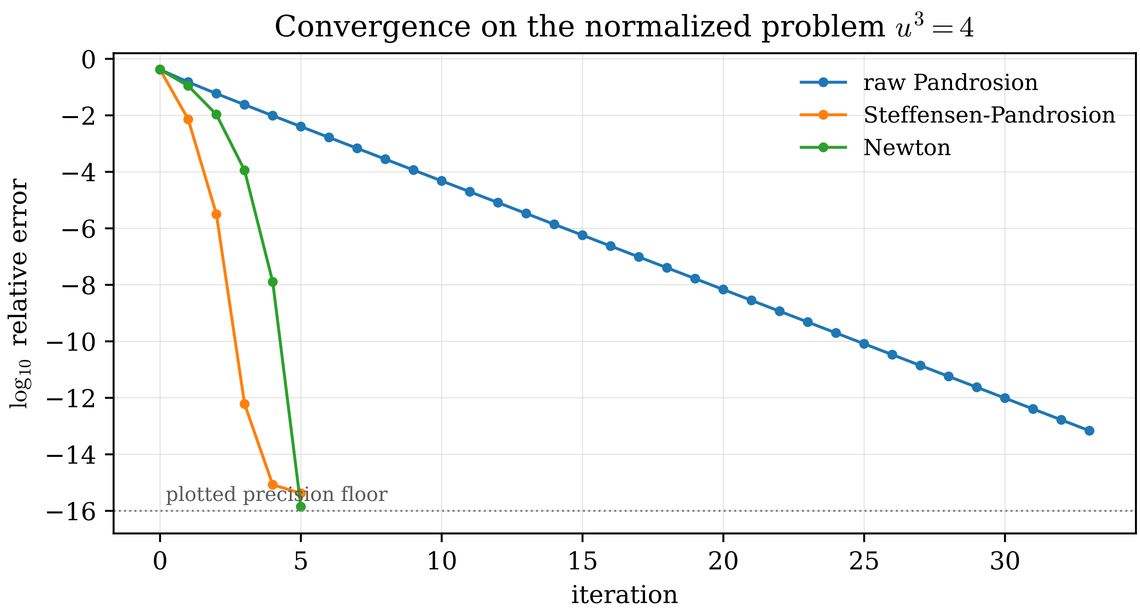

Figure 5: Iteration histories on the normalized problem $u^3 = 4$. Raw Pandrosion is linear, while Steffensen-Pandrosion and Newton show quadratic error collapse; the accelerated curves are truncated once they reach the plotted precision floor.

The Newton column clarifies the intended claim. Newton is dramatically more work-efficient than raw Pandrosion on the unscaled large targets, and it remains a very strong baseline after scaling. The Pandrosion-specific phenomenon is not that it beats Newton as a production root extractor, but that the same geometric map changes from nearly stalled to short and Steffensen-accelerable once the reciprocal homothety makes $|\log y|$ small.

For example, when $p = 3$, $x = 2 \cdot 10^{-6}$, and $B = 50$, the inverted normalized target is

$$y_\infty(B) = \frac{1}{xB^p} = \frac{1}{(2 \cdot 10^{-6})50^3} = 4.$$

Pandrosion therefore works on the small target 4 instead of the large target $1/x = 500000$, and the final answer is returned by

$$z = \frac{1}{50u}.$$

Inversion is a chart change; homothety is a scale change inside that chart; the return map transports the result back to the original variable. The combined procedure accelerates Pandrosion whenever it makes the normalized target closer to 1.

13 Comparison with standard methods

For $f(u) = u^p - x$, Newton's method can be written as

$$u_{n+1} = u_n - \frac{u_n^p - x}{p u_n^{p-1}} = \frac{p-1}{p} u_n + \frac{x}{p u_n^{p-1}}. \tag{24}$$

This is quadratically convergent near a simple root and is very cheap for the monomial problem: one power u_n^{p-1}, one division, and a few scalar operations. Halley's method and the higher Schröder–König families can have even higher local order, at the price of using derivative information or more structured rational formulas; see, for example, Schröder's classical papers [9, 10] and the standard accounts [7, 8].

Relation with Schröder–König iterations. This comparison also sharpens the originality claim. Standard Schröder–König iterations are root-coordinate rational corrections built from f and its derivatives; for a simple root their multiplier at the root is zero when the local order is at least two. Raw Pandrosion has a different local signature. Indeed, in a local branch of the readout map

$$r(s) = x s^{p-1},$$

the induced root-coordinate map is $T = r \circ h \circ r^{-1}$, and therefore

$$T'(x^{1/p}) = h'(s_*) = \lambda_{p,x}.$$

For $x > 1$ this multiplier is generally non-zero. Hence the raw Pandrosion map is not a disguised Newton, Halley, or fixed-order Schröder–König iteration applied to $u^p - x$. Steffensen acceleration later cancels this linear term, but it does so by applying Aitken's Δ^2 process to the Pandrosion fixed-point map, not by replacing it with a derivative-based König formula. This is the precise, limited novelty asserted here: a proportional-geometry map with its own residual multiplier, plus chart scaling that controls that multiplier.

Pandrosion is therefore not advertised here as a faster replacement for Newton on ordinary monomial floating-point computation. Its raw form is linearly convergent, and its Steffensen form is quadratic but requires two evaluations of the fixed-point map h per accelerated update:

$$h(s), \qquad h(h(s)).$$

Each evaluation of h requires the finite geometric sum $S_p(s)$. Evaluated by Horner's rule, this sum costs $O(p)$ scalar operations and one division in the formula for h.

A fair comparison must therefore separate convergence order from work per update. This is also the viewpoint behind the classical Kung–Traub efficiency literature and modern derivative-free methods [11, 12, 13, 14].

Method	Local order	Typical monomial work per update	Role here
Raw Pandrosion	linear	1 evaluation of S_p + 1 division	geometric baseline
Steffensen–Pandrosion	quadratic	2 evaluations of S_p + 2 divisions	derivative-free acceleration
Newton	quadratic	u^{p-1} + 1 division	standard numerical baseline
Halley	cubic	derivatives/rational correction	higher-order baseline

A crude cost-per-digit reading makes the comparison even sharper. Evaluating S_p by Horner's rule costs $O(p)$ scalar operations, while the monomial power u^{p-1} in Newton costs $O(p)$ by a naive product and can be reduced to $O(\log p)$ multiplications by binary exponentiation. Conversely, away from $s = 1$, one may evaluate $S_p(s)$ through $(s^p - 1)/(s - 1)$, which gives a similar fast-power route but introduces an additional division and a removable singularity to handle near $s = 1$. Thus the scalar work model is already favorable to Newton. Even under the conservative convention that one evaluation of S_p and one monomial power evaluation are comparable $O(p)$ blocks, a Newton update roughly doubles the number of correct digits after one such block, whereas one Steffensen–Pandrosion update roughly doubles the number of correct digits after two evaluations of S_p. In this asymptotic efficiency-index sense, Newton has effective index about 2 per monomial block, whereas Steffensen–Pandrosion has effective index about $\sqrt{2}$ per S_p block. Halley is stronger still when its rational formula is available. This is only a crude scalar model, but it prevents the iteration counts from being read as a speed claim.

The work-normalized message is thus modest. For numerical root extraction of a known monomial, Newton is usually the best default. Pandrosion is interesting for a different reason: it gives a proportional-geometry fixed-point operator whose residual multiplier is exactly computable and whose chart-scaled preconditioning is visible in closed form.

Where Pandrosion belongs. The natural niche of this paper is mathematical exposition, historical reconstruction, and geometry-aware preconditioning. The method is useful when one wants to understand how a proportional-mean construction behaves as an iteration, how homothety changes the residual factor, and how reciprocal charts on the Riemann sphere can normalize small targets. It is also a clean testing ground for derivative-free acceleration because all constants are explicit. The intended message is not that Pandrosion dominates Newton. It usually does not. The message is that Pandrosion supplies a geometric fixed-point operator whose residual profile is exactly analyzable and whose performance improves substantially under chart scaling and Steffensen acceleration.

Cyclotomic anchors and local seeding disks ($p = 5$)

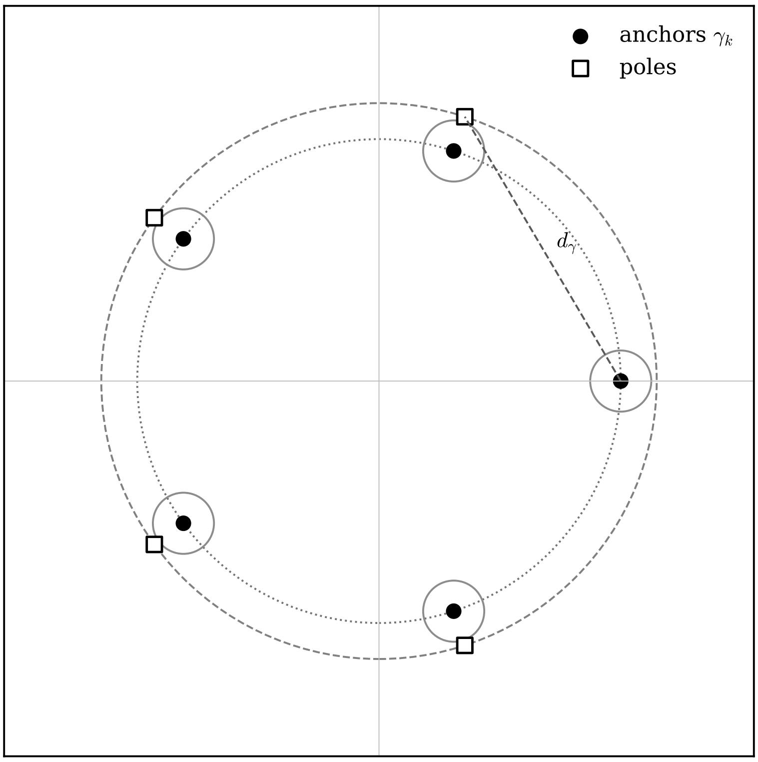

Figure 6: Cyclotomic anchors for the complex monomial problem. The poles are the non-trivial roots of unity; small disks around anchors provide local seeding regions for the accelerated map.

14 Complex extension

The real-line analysis identifies one positive fixed point. Over $\mathbb{C}$, the equation $s^p = 1/x$ has p fixed points:

$$\gamma_k = x^{-1/p}\zeta_p^k, \qquad \zeta_p = e^{2\pi i/p}, \qquad k = 0, \dots, p-1. \tag{25}$$

These are the cyclotomic anchors of the complex Pandrosion map.

The complex dynamics has non-trivial basin structure, as is typical for rational iteration in one complex variable [15]. A practical way to avoid this difficulty is to seed points directly into a known local contraction disk around the desired anchor.

Definition 14.1 (Targeted seed)**.** Given an anchor γ and a parameter $0 < \varepsilon < 1$, define

$$\mathrm{seed}_{\gamma,\varepsilon}(z) = \gamma + \varepsilon(z - \gamma).$$

The seed is a homothety centered at γ. It pulls any point z closer to the target anchor. For ε small enough, the seeded point lies in a local quadratic-convergence region for the Steffensen–Pandrosion map.

Theorem 14.2 (Effective seeded complex convergence)**.** *Let $x \in \mathbb{C} \setminus \{0, 1\}$ and let γ be one of the anchors (25). Then there exists an effectively computable radius $r_\gamma > 0$ such that the Steffensen–Pandrosion map is analytic on $D(\gamma, r_\gamma)$ after filling its removable singularity at γ, fixes γ, and satisfies*

$$|\sigma_{p,x}(z) - \gamma| \le C_\gamma |z - \gamma|^2 \qquad (z \in D(\gamma, r_\gamma))$$

for some computable constant C_γ. Consequently, every seeded point $\mathrm{seed}_{\gamma,\varepsilon}(z)$ with

$$|\,\mathrm{seed}_{\gamma,\varepsilon}(z) - \gamma| < \min\left(r_\gamma, \frac{1}{2C_\gamma}\right)$$

converges quadratically to γ under Steffensen–Pandrosion.

Proof. The poles of $h_{p,x}$ are the zeros of S_p, namely the non-trivial p-th roots of unity. Since $x \ne 1$, the anchor γ is not one of these poles. Put

$$P_p = \{\omega : \omega^p = 1,\ \omega \ne 1\}, \qquad d_0 = \mathrm{dist}(\gamma, P_p) > 0, \qquad R_0 = d_0/4.$$

Because $S_p(z) = \prod_{\omega \in P_p}(z - \omega)$, every $z \in D(\gamma, R_0)$ satisfies

$$|S_p(z)| \ge (3d_0/4)^{p-1}.$$

Together with the finite bounds

$$|S_p'(z)| \le \sum_{k=1}^{p-1} k(|\gamma| + R_0)^{k-1}, \qquad |S_p''(z)| \le \sum_{k=2}^{p-1} k(k-1)(|\gamma| + R_0)^{k-2},$$

this gives explicit bounds for h, h', and h'' on $D(\gamma, R_0)$ from the rational formula for h. In particular one can choose a positive $r_1 \le R_0$ such that $h(D(\gamma, r_1)) \subset D(\gamma, R_0)$.

Now set

$$D(z) = h(h(z)) - 2h(z) + z.$$

The proof of Theorem 9.1 gives the local expansion

$$D(\gamma + e) = (\lambda - 1)^2 e + O(e^2), \qquad \lambda = h'(\gamma),$$

and here $\lambda \neq 1$. Since all derivatives of D on $D(\gamma, r_1)$ are bounded effectively from the preceding rational bounds, one can compute B_γ with

$$|D(\gamma + e) - (\lambda - 1)^2 e| \le B_\gamma |e|^2 \qquad (|e| \le r_1).$$

Choose

$$r_2 = \min\left(r_1, \frac{|\lambda - 1|^2}{2B_\gamma}\right),$$

with the evident convention $|\lambda - 1|^2/(2B_\gamma) = +\infty$ when $B_\gamma = 0$. Then $D(z)$ has no zero in $D(\gamma, r_2) \setminus \{\gamma\}$, so the Steffensen quotient has only its removable singularity at γ. On that disk, fill the singularity by Theorem 9.1 and put

$$C_\gamma = \max_{|z-\gamma|=r_2} \left| \frac{\sigma_{p,x}(z) - \gamma}{(z - \gamma)^2} \right|.$$

This maximum is finite and computable to any prescribed precision because the function is rational and pole-free on the circle. The displayed quadratic estimate follows from the maximum principle. Finally, if $|z_0 - \gamma| < \min(r_2, 1/(2C_\gamma))$, then $|z_{n+1} - \gamma| \le C_\gamma |z_n - \gamma|^2$ keeps the orbit in the disk and forces quadratic convergence. Take $r_\gamma = r_2$. □

Once γ_k is obtained, the corresponding root of x is reconstructed by

$$\rho_k = x\gamma_k^{p-1} = x^{1/p}\zeta_p^{-k}.$$

Thus the complex scheme is a multi-start method indexed by the cyclotomic anchors.

15 Selector regularity

A target selector chooses which anchor should receive a starting point. The simplest selector is the nearest-anchor rule: assign z to the anchor γ_k that minimizes $|z - \gamma_k|$.

This is a finite Voronoi problem. The boundaries between cells are finitely many real line segments or rays determined by perpendicular bisectors of the anchor set. They have planar Lebesgue measure zero. On the complement of these boundaries, the selected anchor is locally constant.

Proposition 15.1 (Piecewise regularity of the monomial selector)**.** *The nearest-anchor selector is Borel measurable on $\mathbb{C}$ and locally constant outside the Voronoi boundary. On any compact set K whose distance from the boundary is at least $\eta > 0$, the selected anchor is constant on every ball of radius $\eta/2$ contained in K. Consequently the seeded initial map*

$$z \longmapsto \gamma + \varepsilon(z - \gamma)$$

is locally ε-Lipschitz on each such cell, and every fixed finite number of Steffensen–Pandrosion updates depends continuously on the input inside the cell.

Proof. Voronoi cells are defined by finitely many inequalities $|z - \gamma_k| \le |z - \gamma_j|$ between continuous functions, hence they are Borel sets. Their boundaries are contained in finitely many real affine lines, so they have measure zero. Away from the boundary, the nearest anchor does not change under sufficiently small perturbations. The seed map has Lipschitz constant ε, and the finite iterates are compositions of rational maps on a pole-free neighborhood, hence are continuous there. □

This regularity statement is intentionally modest. It is not a new measure-theoretic theorem; it only records that the complex multi-start construction has no hidden non-measurable selection step.

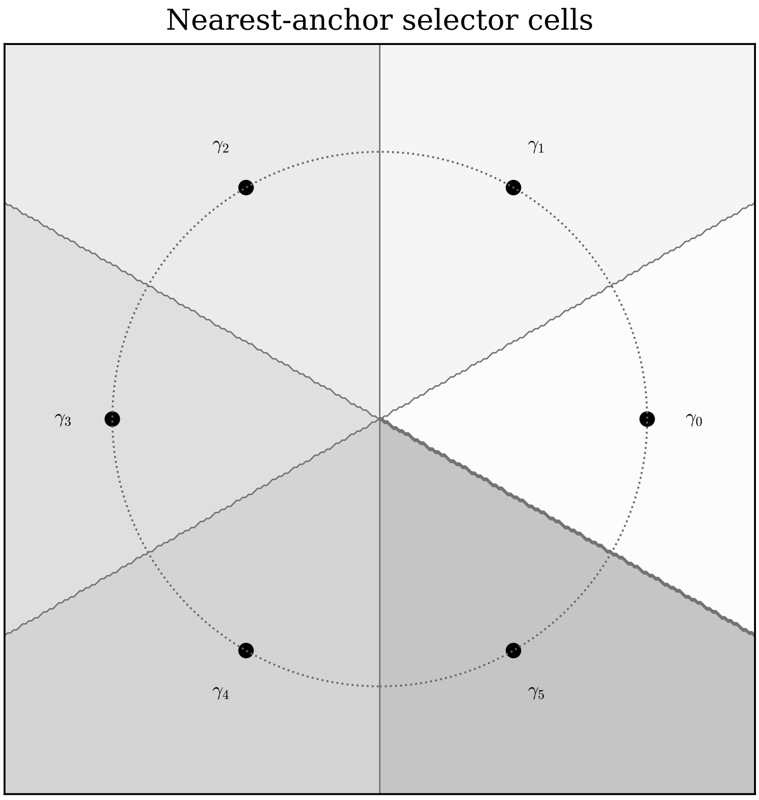

Figure 7: The nearest-anchor selector partitions the complex plane into Voronoi cells. The selector is simple and locally stable away from the cell boundaries.

16 Conclusion

Pandrosion's construction can be read as a monomial root-finding algorithm. In modern notation, it is the fixed-point scheme

$$s_{n+1} = 1 - \frac{x-1}{x(1+s_n+\cdots+s_n^{p-1})}.$$

Its fixed point is $x^{-1/p}$, its readout is $x^{1/p}$, and its contraction ratio is the explicit quantity (7).

The paper's originality is deliberately narrow. It does not claim a new all-purpose competitor to Newton. It claims that the monomial Pandrosion geometry admits a complete residual account organized by six facts:

1. the finite geometric progression of proportional means forces the denominator S_p, and the multiplier is the closed form $1 - p/S_p(\alpha)$;
2. the same multiplier explains both Thales scaling and reciprocal Riemann-chart scaling;
3. the logarithmic rule is local by the expansion $\lambda_{p,y} \sim ((p-1)/(2p)) \log y$ and global on $y \geq 1$ by monotonicity;
4. on a q-lattice, the two-chart rule has the sharp magnitude-independent bound $1-p/S_p(\sqrt{q})$, and K equally spaced offsets are minimax optimal among K-phase palettes;
5. raw Pandrosion is not a standard Schröder–König root-coordinate correction, because in root coordinates its derivative is $T'(x^{1/p}) = \lambda_{p,x} \neq 0$ for $x > 1$;
6. Steffensen acceleration preserves the derivative-free map while making the local convergence quadratic, and the complex monomial picture is handled by cyclotomic anchors and finite anchor selection.

Thus the conceptual contribution is the chart-scaling principle, not raw speed: before asking an iteration to converge, choose the chart in which its residual factor is small. In the Pandrosion viewpoint, homothety and inversion are not post-processing tricks. They are geometric preconditioners for the multiplier of the fixed-point map.

The complex extension adds cyclotomic anchors, finite anchor selection, and seeded local convergence. This light version deliberately centers the monomial equation $z^p = x$, because this is the setting where the finite geometric sum S_p gives a closed-form map and where the residual profile is fully explicit. The elementary algebraic identities used here are suitable for a small formal-audit companion, but no independent proof-assistant artifact is claimed in this manuscript.

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